RV College of Engineering®
Mysore Road, RV Vidyaniketan Post,
Bengaluru - 560059, Karnataka, India

Department of Mechanical Engineering

Date: 06th May 2025	**CIE – 2**	**Max. Marks: 10 + 50**
Semester: IV	**UG**	**Duration: 2 Hrs**
Course Title: Materials Science for Engineers		**Course Code: ME242TB**

S No	Questions	M	BT	CO
	Part A			
1	The stress at which a material begins to deform plastically is called the _____.	1	1	2
2	Piezoelectric materials generate an electric charge in response to applied _____.	1	1	2
3	_____ deformation is fully recoverable when the load is removed.	1	1	2
4	Define optoelectronics with an example.	2	2	2
5	The property of a material that enables it to absorb energy and plastically deform without fracturing is called _____.	1	1	2
6	In which type of semiconductors the electrical behavior is dominated by impurities?	1	1	2
7	_____ are materials used in joint replacements due to their compatibility with biological tissues.	1	1	2
8	Cement and concrete are widely used _____ materials in construction.	1	1	2
9	Fiber-reinforced composites are formed by embedding fibers in a _____ matrix.	1	1	2
	Part B			
1	Explain the working principle of ferroelectricity and piezoelectricity with suitable examples, list their applications.	10	3	2
2	Describe how superconductors and optical fibres work and list their practical applications.	10	2	2
3	Draw and interpret a stress-strain curve for a ductile material.	10	3	2
4	Classify polymers and explain their properties with examples and applications.	10	2	2
5	Write short notes on the following materials: write the composition, grades / types and applications (i) Copper Alloys (ii) Cement (iii) Glass	10	2	2

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Department of Mechanical Engineering

Date: 22nd July 2024	**CIE – II**	**Max. Marks: 50 + 10**
Semester: IV	**UG**	**Duration: 120 Min**
Course Title: Material Science for Engineers		**Course Code: ME242AT**

S No	Questions	M	BT	CO
	Part A			
1	Photodiodes serve as _____ in optical communication systems.	1	1	2
2	A light controlled variable resistor, whose resistance decreases with increasing incident light intensity is known as _____.	1	1	2
3	Name the fundamental semiconductor devices that act as amplifiers or switch.	1	1	2
4	The insulating materials having ability to store and support the transmission of electric charge without conducting it are known as _____.	1	1	2
5	Name the alloying element mainly responsible for the high corrosion resistance property in stainless steel.	1	1	2
6	Mention two advantages of non-ferrous alloys over the ferrous alloys.	2	1	2
7	The process involves shaping a material by forcing it through a die to create a specific cross-sectional shape is known as _____.	1	1	2
8	Name the type of polymer which become soft upon heating and become hard and rigid on cooling also do not have cross linking and branching.	1	1	2
9	Give an example of natural composites.	1	1	2
	Part B			
1	Illustrate the working principle and state the applications of the following optoelectronic devices: i) LED ii) Photo resistor iii) photo transistor.	10	3	2
2a	List any five common dielectric materials along with their applications.	5	2	2
2b	Explain the effects of different alloying elements on steel.	5	3	2
3	Explain the properties and applications of following materials with examples. i) Polymers ii) Metal matrix composites.	10	3	2
4	Illustrate the properties and applications of four types of cast iron.	10	3	2
5a	Explain the metal casting process. Write the advantages and disadvantages of any two casting processes.	5	3	2
5b	Write the properties and applications of bio-materials.	5	2	2

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Department of Biotechnology

Date: May 6th 2025	**Closed Book Offline Test – 2**	**Max. Marks: 60**
Semester: IV	**UG**	**Duration: 120 Min**
Course Title: Bio Safety Standards and Ethics (Basket Course)		**Course Code: BT242TC**

S No	Part A – Questions (Quiz)	M	BT	CO
1	HACCP is a joint management system developed by _____ and _____.	2	2	3
2	A man found a copper wire in pizza and human finger in ice cream. Mention the different types of the hazards.	2	2	3
3	Mention the 2-hour and 4-hour rule for food storage.	2	2	3
4	Decode the fssai license number of campa cola (Lic. No. 10012022000857).	2	2	4
5	Expand the term GAP and GHP give its importance.	2	2	4

S No	Part B – Questions (Test)	M	BT	CO
1–3	In the wheat processing from farm to customer, the following steps are involved. (ANSWER Q1, Q2, Q3) a) Farmers spend time planting seeds, checking for disease and monitoring plant health until harvest. Combines harvest the wheat kernels, which are then loaded into a semi-truck. b) The harvested grain is sold at market price to a local grain elevator. The elevator can store the grain until the right market price, or it can sell it. c) Country elevators sell their grain to terminal elevators, which clean, separate and maintain the value of the grain. The grain is then sold to flour millers for domestic consumption, or it is loaded into ships bound for overseas markets. d) The flour mills grind the grain into different types of flour — whole wheat, all-purpose, bread flour, etc. The mills can also use the flour to make ready-to-eat products. Both the flour and wheat products are shipped to grocery stores and other food retailers. e) Bakers and chefs also use flour to make a wide variety of delicacies. Consumers can either buy these products, or buy flour produced by the flour mills to make their own creations at home. Finally, wheat products make their way to your table. These products are essential to many holidays and celebrations and go through a safe process to get to you. Whenever you purchase breads, buns, pizza, rotis think of a wheat farmer and all of those in between.			
1	Identify and elaborate on the type of hazard in each step of the wheat processing.	7	1	3
2	Mention and justify the elimination method for identified hazard during the wheat processing.	6	2	3
3	How we can implement the HACCP for the wheat processing.	7	2	3
4	Illustrate on various Techniques that take place in a food testing laboratory.	10	3	3
5	Elaborate on food borne diseases and food allergens.	10	3	4
6	Justify GMP and GLP in food processing.	10	4	3

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Department of Biotechnology

Date: July 23, 2024	**Closed Book Offline Quiz and Test – 2**	**Max. Marks: 10 + 50**
Semester: IV	**UG**	**Duration: 120 Min**
Course Title: Bio Safety Standards and Ethics (Basket Course)		**Course Code: BT242AT**

S No	Questions	M	BT	CO
	Part A			
1	In food, the term used for disease causing microorganism is referred as _____.	1	2	3
2	Mention the temperature range which is optimum for bacterial growth in food.	1	1	4
3	What is the 6th principle of HACCP?	1	2	3
4	Mention the instrument used to multiply the DNA.	1	1	3
5	Mention the concentration of the lead in food as per the limit for lead in food as per FSSAI.	1	2	3
6	Mention any two food certification followed in the world.	1	2	3
7	Expand the term GAP.	1	2	4
8	People, premises, processes, products and procedures is related to _____.	1	3	4
9	Mention any two examples for microbiological contamination.	1	2	3
10	Mention the number of Licenses issued by FSSAI.	1	2	3
	Part B			
1	Mention the physical, chemical, and biological hazardous material which contaminate the food.	10	2	3
2	Expand the term FSSAI, illustrate its functions, importance & type of license issued by them.	10	3	3
3	Discuss various methods to eliminate the chemical, Biological and physical hazardous materials from food.	10	3	3
4	Expand the term HACCP and explain in detail the principles of HACCP.	10	3	3
5a	List out the food spoilage microorganisms and explain the diseases caused by them.	5	3	4
5b	Mention the GMP and GLP in maintaining the food hygiene.	5	3	4

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Department of Civil Engineering

Date: 6.5.25	**Quiz 2 and Test – 2**	**Max. Marks: 10 + 50**
Semester: IV	**UG**	**Duration: $1\frac{1}{2}$ Hrs**
Course Title: Environment and Sustainability		**Course Code: CV242TA**

S No	Questions	M	BT	CO
	Quiz			
1.	Highlight the principle of green engineering.	2	1	2
2.	Enumerate 10% rule of Energy flow.	2	1	2
3.	Hydrogen fuel cells produce electricity by combining _____ and _____ atoms.	2	1	2
4.	List two renewable and non-renewable sources of energy.	2	1	1
5.	Define carbon sequestration.	2	1	2
	Test			
1.	Enumerate the process of energy management and energy conservation with its objectives.	10	2	2
2.	Demonstrate on “Hydrogen energy is a clean energy” write the advantages and Disadvantages of Hydrogen energy as a energy source.	10	2	2
3.	Explain (I) Biomass energy (II) Geothermal Energy.	10	2	1
4.	With a neat sketch explain carbon cycle and enumerate the steps involved in the same.	10	2	2
5.	Justify how renewable energy benefit environment and save money.	10	3	4

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Department of Civil Engineering

Date: 23/7/2024	**CIE – II**	**Max. Marks: 50**
Semester: IV	**UG**	**Duration: 90 Min**
Course Title: Environment and Sustainability		**Course Code: CV242TA**

S No	Questions	M	BT	CO
	Note: All questions are compulsory.			
1.	The last two decades shows an alarming increase in global temperatures. The rise has been directly linked to the increase in _____.	2	1	2
2.	Write two disadvantages of Tidal energy.	2	1	3
3.	What is meant by Carbon Sequestration.	2	1	1
4.	Write two objectives of fair trade.	2	1	3
5.	How Corporate Social Responsibility leads to sustainability.	2	2	3
	Test – I			
1.	Brief on i) Bio mass energy ii) Ocean Thermal Energy conversion.	10	1	1
2.	Demonstrate on “Hydrogen energy is a clean energy” write the advantages and Disadvantages of Hydrogen energy as a energy source.	10	3	3
3.	Write on four approaches and principles of Green engineering.	10	1	3
4.	Explain the importance of Environmental auditing towards the sustainable growth.	10	2	2
5a	Explain Source and sink in Carbon cycle.	5	2	1
5b	Explain the prescriptive based and incentive based Environmental Economics.	5	2	3